

Normalization & Dependency Preservation

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Dependency Preservation?

Definition

A decomposition is Dependency Preserving if all Functional Dependencies (FDs) from the original relation can still be enforced without joining the decomposed tables.

Simple Definition

After splitting a table, you should still be able to enforce every original rule directly on the smaller tables. No joins should be required.

Dependency Preservation allows:

- Easier constraint checking
- Faster updates
- Simpler application logic
- Better performance.

Preservation Tests

Method 1 (Interview-Friendly)

Check every original FD. Ask: Can this FD be checked inside one decomposed relation? If yes, Dependency Preserved.

BCNF Trade-offs

One of the most important interview concepts.

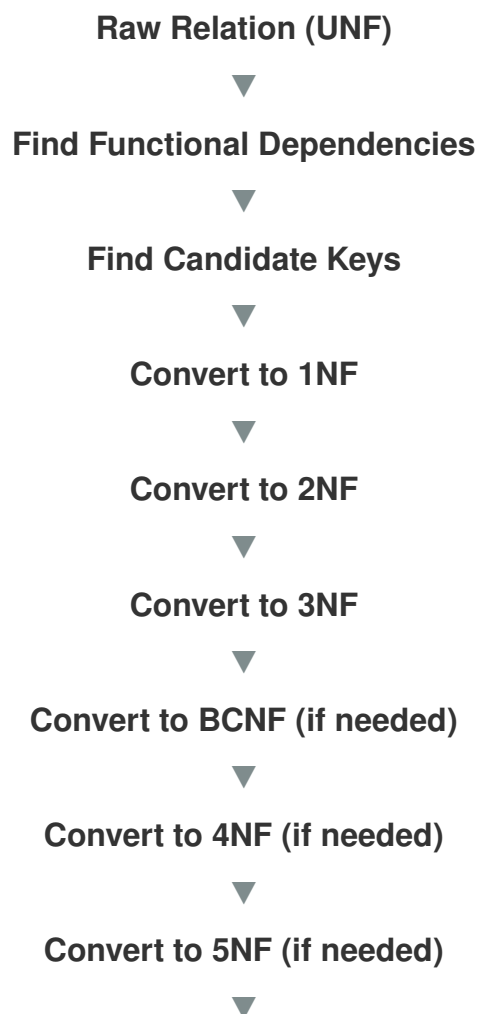
3NF generally provides:

- Lossless Join
- Dependency Preservation

BCNF guarantees:

- Lossless Join
- But may lose Dependency Preservation.

Complete Normalization Algorithm



Verify Lossless Join



Verify Dependency Preservation



Final Database Schema